GRIN1 Hypothesis Deep Research — GluN1 Glycine-Site Occupancy Auto-Inhibits GluN1/GluN3 Current

Gene: GRIN1 (GluN1 / NR1) · **UniProt:** Q05586 · **Organism:** *Homo sapiens* (NCBITaxon:9606) **Term under review:** glycine-gated cation channel activity (GO:0160212) **Hypothesis slug:** `glun1-glycine-site-inhibits-glun1-glun3-current`

Summary

Verdict: SUPPORTED. In diheteromeric GluN1/GluN3A and GluN1/GluN3B glycine-gated receptors ("excitatory glycine receptors," eGlyRs), occupancy of the **GluN1 (GRIN1) glycine-binding site is a negative regulator** of the glycine-evoked current. Glycine binding to GluN1 drives auto-inhibition and promotes desensitization of the maximal current (Imax), whereas glycine binding to the **GluN3 subunit is the event that opens the channel**. This is the mechanistic *opposite* of GluN1's role in conventional GluN1/GluN2 NMDA receptors, where GluN1 glycine-site occupancy is an obligatory *positive* co-agonist event. The seed hypothesis is directly affirmed by the primary literature, including all four reference PMIDs supplied in the source context, with no contradicting study identified.

The evidence base is convergent and spans multiple experimental paradigms and both recombinant and native receptors. Pharmacologically, GluN1 glycine-site **antagonists** (CGP-78608, MDL-29951-class competitive antagonists, 7-chlorokynurenate) and GluN1 ligand-binding-domain (LBD) **mutations** that disrupt glycine binding *potentiate* GluN1/GluN3A currents by roughly 10- to 30-fold, converting small, rapidly desensitizing responses into large, stable responses. Conversely, antagonism or mutation of the **GluN3** glycine site *reduces or abolishes* activation. Structural work by cryo-EM (2024–2026) shows that channel gating is driven solely by GluN3A conformational changes, while the glycine-bound/CGP-antagonized state of GluN1 governs entry into (or escape from) the desensitized state. These findings have been reproduced across Xenopus oocytes, HEK293 cells, and native mouse hippocampal neurons.

For GO curation, the specific term under review — GO:0160212 "glycine-gated cation channel activity" on Q05586 — is currently annotated by **ISS (inferred from sequence similarity)**, not by direct experiment. The hypothesis does not argue that GRIN1 lacks a role in a glycine-gated channel; it argues about the *sign* of GluN1 glycine-site occupancy. The channel activity is a property of the **assembled GluN1/GluN3 complex**, and the gating glycine sensor is GluN3, not GluN1. The recommended curation lead is therefore to **retain GO:0160212 as a complex-level / `contributes_to` qualified annotation**, and to ensure GRIN1 is **not** annotated as the activating glycine sensor for this receptor class; a complementary negative-regulation/desensitization annotation better captures GluN1's actual contribution.

Key Findings

Finding 1 — GluN1 glycine-site occupancy auto-inhibits GluN1/GluN3 current; GluN3 glycine site gates the channel

The central claim of the hypothesis is directly stated and experimentally demonstrated in the primary electrophysiology literature. Madry et al. (2007), in a study explicitly on the "principal role of NR3 subunits," reported that **"glycine bound to the NR1 subunit causes auto-inhibition of NR1/NR3 receptors whereas glycine binding to the NR3 subunits is required for opening of the ion channel"** (PMID: 17214961). This is a verbatim statement of the seed hypothesis. The same study provided the quantitative backbone: **"antagonists of and substitutions within the glycine-binding site of NR1 potentiate NR1/NR3 receptor function up to 25-fold, but inhibition or mutation of the NR3 glycine binding site reduces or abolishes receptor activation."** The two glycine sites therefore carry *opposite signs*: disabling GluN1's site potentiates, while disabling GluN3's site abolishes.

This dissociation is reinforced by a compound-level result in native and recombinant systems. Grand et al. (2018) identified **CGP-78608**, a GluN1 glycine-site antagonist, which **"greatly enhances GluN1/GluN3A responses, converting small and rapidly desensitizing currents into large and stable responses"** (PMID: 30425244). Critically, the potentiating effect of blocking the GluN1 site was observed not only in recombinant systems but also in **native P8–P12 mouse hippocampal neurons**, establishing physiological relevance beyond artificial expression systems.

The structural capstone comes from cryo-EM across the receptor's functional states. Kim/Gouaux and colleagues (2026) showed that **"glycine activates the receptor solely via GluN3A-dependent conformational changes, opening the gate with two-fold symmetry, and induces a roughly four-fold symmetric desensitized state. CGP-bound GluN1 restricts GluN3A rotation, promoting glycine-induced activation by blocking desensitization"** (PMID: 42665662). This provides a physical mechanism: gating is a GluN3A motion, and GluN1 glycine occupancy biases the assembly toward the desensitized (non-conducting) conformation; antagonizing GluN1 (CGP) relieves that bias.

Finding 2 — All four seed reference PMIDs independently confirm GluN1 glycine-site auto-inhibition

Each of the four references in the source context was resolved and independently supports the negative sign, with no contradiction:

- **Mesic et al. 2016 (Neuropharmacology)** states verbatim: **"Binding of glycine to the high-affinity GluN3 subunits triggers channel opening, whereas glycine binding to the low-affinity GluN1 subunits causes an auto-inhibition of the maximal glycine-inducible receptor current (Imax). Hence, competitive antagonists of the GluN1 subunit strongly potentiate glycine responses of wild type (wt) GluN1/GluN3 receptors."** (PMID: 26777280). This is the most explicit statement of the hypothesis and adds the mechanistic nuance that the two glycine sites differ in affinity (GluN3 high-affinity / activating; GluN1 low-affinity / inhibitory).
- **Awobuluyi et al. 2007 (Mol Pharmacol)** adds a mechanistic refinement: **"the apo conformation of NR1 in NR1/NR3 receptors is permissive for receptor activation. Agonist-bound NR1 may also contribute to peak NR1/NR3 receptor currents but specifically enables significant NR1/NR3 receptor current decay under the conditions studied here, presumably via a slow component of desensitization"** (PMID: 17047094). This clarifies that the *apo* (unliganded) GluN1 state is permissive for opening, and that liganded GluN1 principally drives current decay (desensitization) — consistent with the negative sign, while noting a context-dependent contribution to peak current.
- **Michalski & Furukawa 2024 (Sci Adv)** provides structural support: cryo-EM of GluN1/GluN3A with the antagonist CNQX versus glycine showed distinct GluN3A orientations between states, and **"Site-directed disruption of the unique subunit interface in the glycine-bound structure mitigated desensitization"** (PMID: 38598639). This directly links the glycine-bound conformation to desensitization, i.e., glycine occupancy promotes the inhibited state.

- **Chatterton et al. 2002 (Nature)** is the founding description: NR1/NR3A or NR1/NR3B co-assemble into excitatory glycine receptors "unaffected by glutamate or NMDA, and **inhibited by D-serine**" (PMID: 11823786). Because D-serine is a GluN1 glycine-site agonist, its inhibitory action on NR1/NR3 is early, direct evidence that occupancy of the GluN1 site is negative for current.

Finding 3 — Independent replications and the GRIN1 F484 glycine-pocket residue

The negative sign has been independently replicated beyond the Madry/Mesic lineage. Smothers & Woodward (2009) reported that disrupting GluN1 glycine binding **"produced a greater than 30-fold potentiation of currents during coapplication of glycine and the glycine antagonist 7-chlorokynurenic acid"** in HEK293 cells, and that the effect was NR1 splice-variant-dependent (PMID: 19726695). Madry et al. (2008) separately showed ~10-fold potentiation by an NR1 antagonist and supralinear (>120-fold) potentiation when combined with Zn^{2+} (PMID: 18711142). The consistency of the direction (potentiation upon GluN1-site blockade) across three independent laboratories and expression systems is a strong convergence signal.

On the molecular anatomy relevant to the hypothesis's cited **F484A** LBD mutant: a direct UniProt REST query of Q05586 (938 aa) confirmed that **residue 484 is a phenylalanine (F)** located in the S1 glycine-binding lobe (sequence context 478-`LVADGKFGTQERVNN`-492). This confirms that the F484A mutation referenced in the hypothesis maps to the human GluN1 glycine-binding pocket, consistent with the experiments that use GluN1 LBD substitutions to disrupt glycine binding and thereby potentiate GluN1/GluN3 current.

The same UniProt record shows that the GO:0160212 annotation on Q05586 carries evidence code **ISS:UniProt** (inferred from sequence similarity), whereas the experimentally supported GO terms on the record include GO:0016594 "glycine binding" (IDA), GO:0017146 "NMDA selective glutamate receptor complex" (IDA), and GO:1905429 "response to glycine" (IDA). The term under curation is therefore an electronic/similarity inference rather than a direct experimental annotation.

Mechanistic Model / Interpretation

The two classes of GluN1-containing NMDA-receptor-family channels invert the role of the GluN1 glycine site:

	GluN1/GluN2 NMDAR (conventional)	GluN1/GluN3 eGlyR (excitatory glycine receptor)
Subunits	2x GluN1 + 2x GluN2	2x GluN1 + 2x GluN3
Gating agonist(s)	glutamate (GluN2) + glycine (GluN1) — BOTH required	glycine ONLY (no glutamate site)
GluN1 glycine site	POSITIVE (obligatory co-agonist; occupancy REQUIRED to open)	NEGATIVE (occupancy drives auto-inhibition / desensitization)
GluN3 glycine site	n/a	POSITIVE (occupancy OPENS the gate)
Effect of blocking the GluN1 site	Inhibits / closes channel	POTENTIATES current ~10–30x (CGP-78608, 7-Cl-kynurenate, F484-pocket mutants)
Effect of blocking the GluN3 site	n/a	ABOLISHES / reduces activation

A state-based reading reconciles the apparent nuance in Awobuluyi 2007:

```
  apo-GluN1 + glycine-GluN3   --->   OPEN (conducting)      [GluN1 apo is permissive]
                                          |
  glycine binds GluN1          -------->  promotes rotation toward
                                         DESENSITIZED (non-conducting, ~4-fold symmetric)
                                          |
  CGP-78608 bound to GluN1     -------->  restricts GluN3A rotation,
                                         BLOCKS desensitization --> large, stable current
```

In this model, glycine acting at the high-affinity GluN3 site is the gating trigger. Glycine at the lower-affinity GluN1 site is dispensable for opening (apo-GluN1 is permissive) and, when occupied, destabilizes the open state in favor of desensitization, lowering Imax. The GluN1 glycine-site antagonist CGP-78608 clamps the GluN1 LBD in an antagonist-bound conformation that sterically/allosterically prevents the GluN3A rotation underlying desensitization, yielding sustained current. This is a clean, structurally grounded mechanism that unifies the pharmacology (potentiation by GluN1 antagonists), the mutagenesis (potentiation by GluN1 glycine-pocket mutants such as those in the F484 region), and the cryo-EM state assignments.

Evidence Matrix

Citation	Evidence type	Stance	Claim tested	Key finding	Context	Co lir
PMID: 17214961 Madry 2007	Direct assay + mutant	**Supports**	Sign of GluN1 vs GluN3 glycine occupancy	GluN1-bound glycine = auto-inhibition; GluN3 glycine = opening; GluN1 antagonists/ mutants potentiate up to 25×, GluN3-site disruption abolishes activation	Recombinant NR1/NR3, Xenopus oocytes	Hi re
PMID: 26777280 Mesic 2016	Direct assay + mutant	**Supports**	Imax auto-inhibition via GluN1	High-affinity GluN3 triggers opening; low-affinity GluN1 glycine causes auto-inhibition of Imax; GluN1 competitive antagonists strongly potentiate	Recombinant, Xenopus oocytes	Hi re
PMID: 17047094 Awobuluyi 2007	Direct assay + mutant	**Supports (qualifies)**	Role of apo vs agonist-bound GluN1	Apo-GluN1 permissive for activation; agonist-bound GluN1 enables current decay (slow desensitization); may contribute to peak	Recombinant, oocytes	Hi nu pe co
PMID: 38598639	Structural (cryo-EM)	**Supports**		Glycine-bound state linked to	Recombinant GluN1/	

Citation	Evidence type	Stance	Claim tested	Key finding	Context	Co lin
Michalski & Furukawa 2024			Structural basis of desensitization	desensitization; disrupting the glycine-bound subunit interface mitigates desensitization; 1-3-1-3 arrangement	GluN3A, cryo-EM	Hi str vi
PMID: 30425244 Grand 2018	Direct assay (pharmacology)	**Supports**	GluN1 antagonist potentiates	CGP-78608 converts small/ desensitizing GluN1/GluN3A currents into large/stable responses	Recombinant + **native** P8–P12 mouse hippocampal neurons	Hi na re
PMID: 42665662 Kim/ Gouaux 2026	Structural (cryo-EM)	**Supports**	Which subunit gates; GluN1 role in desensitization	Gating = GluN3A conformational change; CGP-bound GluN1 restricts GluN3A rotation and blocks desensitization	Recombinant GluN1-GluN3A, cryo-EM states	Hi str
PMID: 19726695 Smothers & Woodward 2009	Direct assay + mutant	**Supports**	GluN1-site disruption potentiates	>30× potentiation with glycine + 7-chlorokynurenate when GluN1 glycine binding disrupted; NR1 splice-variant-dependent	Recombinant, HEK293	Hi in re
PMID: 18711142 Madry 2008	Direct assay	**Supports**	GluN1 antagonist + Zn^{2+} potentiation	~10× potentiation by NR1 antagonist; supralinear	Recombinant, oocytes	Hi in re

Citation	Evidence type	Stance	Claim tested	Key finding	Context	Co lin
				>120× with Zn^{2+}; effect mapped to GluN1 LBD		
PMID: 11823786 Chatterton 2002	Direct assay	**Supports**	GluN1-site agonist inhibits	NR1/NR3 glycine-activated; **inhibited by D-serine** (GluN1-site agonist)	Recombinant + cortical neurons	Hi fo st
PMID: 32389749 EU1180-438 2020	Direct assay	**Qualifies**	Allosteric (non-glycine-site) negative modulation	Selective NAM of GluN1/GluN3; active at native GluN1/GluN3A in CA1 neurons	Recombinant + native hippocampus	Or ph no sit
UniProt Q05586 (computational)	Computational / database	**Context**	Residue identity & annotation status	F484 confirmed in GluN1 S1 glycine lobe; GO:0160212 = ISS (not experimental)	REST query	Di da

No study in the reviewed literature (32 papers) contradicts the negative sign of GluN1 glycine-site occupancy for GluN1/GluN3 current.

Evidence Base

The supporting literature forms three mutually reinforcing layers:

1. **Pharmacology/electrophysiology (functional sign).** Madry 2007 (PMID: 17214961), Mesic 2016 (PMID: 26777280), Madry 2008 (PMID: 18711142), Smothers & Woodward 2009 (PMID: 19726695), and Grand 2018 (PMID: 30425244) collectively establish that blocking or mutating the GluN1 glycine site potentiates current (~10–30×, up to >120× with Zn^{2+}), while blocking the GluN3 glycine site abolishes activation. This is the strongest, most direct evidence.

2. **Structural biology (mechanism).** Michalski & Furukawa 2024 (PMID: 38598639) and Kim/Gouaux 2026 (PMID: 42665662) show the gating motion is GluN3A-driven and that the glycine-bound GluN1 state promotes the desensitized conformation, which CGP-78608 blocks. Native-architecture cryo-EM (PMID: 42039402) further corroborates the gating model.
3. **Founding/receptor-identity work and reviews (orientation).** Chatterton 2002 (PMID: 11823786) defined NR1/NR3 as excitatory glycine receptors inhibited by D-serine. Reviews (e.g., PMID: 37248111, PMID: 27558536) synthesize eGlyR biology and physiological roles (e.g., ventral hippocampus, medial habenula, anxiety behaviors — PMID: 40924452, PMID: 31601771), providing in vivo relevance but used here only as orientation.

GO Curation Implications

The term under review: GO:0160212 "glycine-gated cation channel activity" (MF), currently on Q05586 with evidence code **ISS:UniProt**.

Lead (requires curator verification):

- **Retain GO:0160212 as a complex-level, qualified annotation.** The glycine-gated cation channel is a genuine activity of the assembled GluN1/GluN3 receptor, and GluN1 is an obligatory structural subunit of that complex. However, the *gating glycine sensor* is GluN3, not GluN1. The most accurate representation annotates GRIN1 with a `contributes_to` qualifier on GO:0160212, acknowledging that the activity belongs to the complex while GluN1 is not the activating ligand sensor.
- **Do not annotate GRIN1 as the positive/activating glycine sensor for this receptor class.** GluN1 glycine-site occupancy is negative (auto-inhibition/desensitization) for GluN1/GluN3 current. An unqualified MF annotation risks implying the GluN1 glycine site drives opening, which is false for GluN1/GluN3 (and opposite to its true role there, though positive in GluN1/GluN2).
- **Consider complementary negative-regulation / desensitization annotations.** GluN1's actual contribution to GluN1/GluN3 physiology is captured by negative regulation of channel activity / promotion of receptor desensitization. A BP term such as "negative regulation of monoatomic cation transmembrane transport" or a desensitization-related term, annotated with `contributes_to`, would more faithfully represent the experimental data (Madry 2007; Mesic 2016; Michalski & Furukawa 2024).

- **Experimental upgrade opportunity.** GO:0160212 is currently ISS. Direct experimental evidence (IDA/IMP) for the complex-level activity exists and could support upgrading the evidence code for the *complex* while keeping the sign caveat for the GluN1 subunit.

This is an **MF**-centered decision with a **BP** complement. "Protein binding" is not an appropriate fallback here — the data support specific channel-activity and regulatory terms.

Mechanistic Scope

The hypothesis concerns an **immediate molecular function**: the effect of ligand occupancy at the GluN1 glycine-binding pocket on ion flux through the assembled GluN1/GluN3 channel. The evidence addresses this directly via (i) competitive antagonists at the GluN1 glycine site (CGP-78608, 7-chlorokynurenate), (ii) point mutations/substitutions in the GluN1 LBD glycine pocket (including the F484-containing S1 lobe), and (iii) cryo-EM conformational state assignments. These are genuine measurements of channel current and receptor conformation — not downstream phenotypes.

Downstream/contextual phenomena that should be kept **separate** from this molecular-function claim include: the physiological roles of eGlyRs in ventral hippocampus plasticity and anxiety (PMID: 40924452), aversion control in the medial habenula (PMID: 31601771), myelin signaling (PMID: 20739572), and non-neuronal contexts (erythroid cells, chondrocytes, heart). These establish in vivo relevance but are not evidence about the *sign* of GluN1 glycine-site occupancy per se.

Conflicts and Alternatives

- **Peak-vs-decay nuance (not a true conflict).** Awobuluyi 2007 (PMID: 17047094) notes that agonist-bound GluN1 "may also contribute to peak" current while principally enabling decay/desensitization. This refines, but does not reverse, the negative sign: apo-GluN1 is permissive for opening, and the dominant effect of GluN1 occupancy is desensitization/Imax reduction. Curators should note that "negative" here means suppression of maximal/sustained current, with a possible small positive contribution to transient peak under some conditions.

- **Paralog/subtype context.** The sign is *opposite* in GluN1/GluN2 NMDARs, where GluN1 glycine occupancy is an obligatory positive co-agonist. Any annotation must be subtype-qualified so that the GluN1/GluN3-specific negative role is not generalized to all GluN1-containing receptors, and vice versa.
- **Organism/system generality.** Most mechanistic data are recombinant (Xenopus oocytes, HEK293). Native confirmation exists for the *potentiation-by-GluN1-antagonist* phenomenon (Grand 2018, mouse hippocampal neurons; EU1180-438 at native CA1 GluN1/GluN3A). A caveat: diheteromeric GluN1/GluN3 receptors were historically hard to detect at the surface of mammalian cells (PMID: 18779583), so quantitative native parameters remain less characterized than recombinant ones.
- **No contradicting study was found** across the 32 papers reviewed.

Limitations and Knowledge Gaps

1. **Recombinant dominance of mechanistic detail.** The sign and magnitude (10–30×) of GluN1-site effects are best quantified in recombinant systems. *What was checked:* functional and structural primary literature. *Why it matters:* native receptor stoichiometry, triheteromers, and auxiliary factors could modulate the quantitative effect. *Resolution:* systematic native recordings (e.g., CGP-78608 potentiation) across brain regions and developmental stages.
2. **Peak contribution of GluN1 occupancy.** Awobuluyi's observation that agonist-bound GluN1 may contribute to peak current is condition-dependent. *Why it matters:* a blanket "negative" statement slightly oversimplifies. *Resolution:* kinetic modeling / single-channel analysis isolating peak vs steady-state components as a function of GluN1-site occupancy.
3. **F484A-specific data.** The hypothesis cites GluN1 F484A specifically; the knowledge base confirms F484 is in the human GluN1 S1 glycine lobe, but the strongest functional data use a range of GluN1 glycine-pocket perturbations rather than F484A uniquely in the human orthologue. *Resolution:* confirm the exact F484A current phenotype in a primary source and map to human numbering.
4. **GO structure for complex-level, subunit-divergent activity.** GO annotation of a channel activity whose ligand sensor is a *different* subunit than the one being annotated is inherently awkward. *Why it matters:* the sign caveat is easily lost in a bare MF annotation. *Resolution:* adopt `contributes_to` qualifiers and a paired negative-regulation BP term.

Discriminating Tests

The hypothesis is already well supported; the following would most efficiently close residual gaps or stress-test it:

1. **Occupancy-clamped electrophysiology:** Titrate a GluN1-selective glycine-site agonist (e.g., D-serine) against a GluN1-selective antagonist (CGP-78608) at fixed GluN3-site saturation, measuring Imax and desensitization time constants — directly maps the sign and magnitude of GluN1 occupancy vs current.
2. **F484A (human numbering) functional test:** Express human GluN1-F484A / GluN3A and confirm potentiation and loss of desensitization relative to wild type, nailing down the exact residue cited in the hypothesis.
3. **State-resolved structure–function coupling:** Pair cryo-EM desensitized-state capture (glycine-bound GluN1) with interface-disrupting mutations (as in Michalski & Furukawa 2024) and single-channel recordings to quantify how GluN1 occupancy shifts open-probability vs desensitization.
4. **Native-receptor pharmacology panel:** Apply CGP-78608 and EU1180-438 to native GluN1/GluN3A currents across ventral hippocampus, medial habenula, and myelin to confirm the recombinant sign in situ.

Proposed Follow-up Actions / Curation Leads (require curator verification)

Candidate references with exact snippets to verify:

- PMID: 17214961: *"glycine bound to the NR1 subunit causes auto-inhibition of NR1/NR3 receptors whereas glycine binding to the NR3 subunits is required for opening of the ion channel"* — core negative-sign statement.
- PMID: 26777280: *"glycine binding to the low-affinity GluN1 subunits causes an auto-inhibition of the maximal glycine-inducible receptor current (Imax). Hence, competitive antagonists of the GluN1 subunit strongly potentiate..."*

- PMID: 30425244: *"CGP-78608 … converting small and rapidly desensitizing currents into large and stable responses"* — antagonist potentiation; native + recombinant.
- PMID: 42665662: *"glycine activates the receptor solely via GluN3A-dependent conformational changes … CGP-bound GluN1 restricts GluN3A rotation, promoting glycine-induced activation by blocking desensitization"* — structural mechanism.
- PMID: 17047094: *"the apo conformation of NR1 … is permissive for receptor activation. Agonist-bound NR1 … enables significant NR1/NR3 receptor current decay … via a slow component of desensitization"* — peak-vs-decay nuance.

Candidate GO actions:

Term	Current	Lead action
GO:0160212 glycine-gated cation channel activity (MF)	ISS, unqualified	**Retain** but add `contributes_to` qualifier (complex-level); do not imply GluN1 is the gating glycine sensor
Negative regulation of cation transmembrane transport / desensitization (BP)	absent	**Add** as `contributes_to` to capture GluN1's auto-inhibitory role
GO:0016594 glycine binding (MF)	IDA	Retain (experimental; accurate for GluN1)

Suggested curator questions: 1. Should a subunit whose glycine site *negatively* regulates a complex-level channel activity carry the channel-activity MF term, or only a regulation term plus `contributes_to`? 2. Can GO:0160212 be upgraded from ISS to an experimental code at the complex level while preserving the subunit-level sign caveat?

Suggested experiments: human GluN1-F484A functional confirmation; CGP-78608/EU1180-438 native-tissue panel (see Discriminating Tests).

Bottom Line

The seed hypothesis is **SUPPORTED** by convergent primary evidence (pharmacology, mutagenesis, structure) from recombinant and native receptors, including all four reference PMIDs, with no contradicting study. GluN1 glycine-site occupancy is a **negative** regulator of GluN1/GluN3 current (auto-inhibition/desensitization); GluN3 glycine binding opens the channel — the inverse of GluN1's positive co-agonist role in GluN1/GluN2 NMDARs. For

curation, GO:0160212 on GRIN1 is currently ISS and should be retained as a **qualified, complex-level** annotation (`contributes_to`), paired with a negative-regulation/desensitization annotation, and must not portray GluN1 as the activating glycine sensor for this receptor class.
